Examiner reports

Q1.

This question was answered very well with most candidates gaining nearly full marks. In part (a) many wrote £80 instead of just 80 but this was accepted.

Part (b) was mostly done accurately with many getting to $k = \sqrt[56]{62.5}$ which was accepted as *showing that*. Those who used logarithms lost a mark if they didn't show how log62.5 = 561og$k$: led to the given value of k.

Virtually all candidates got part (c) correct, with both the approximate answer 200707 and the more exact 200648 being seen, often with several decimal places attached, but this was of no matter as candidates had demonstrated a value greater than £200,000. In part (c)(ii) there were occasional errors although most candidates used logarithms to derive 124.7 correctly whilst a few used trial and improvement to deduce 125. However, many lost the last mark in incorrect interpretation of the year, or through not actually giving a year. The year 2025 was the common wrong answer.

Q2.

This question on logarithms was answered better than questions on the same topic last year although this area of the specification continues to cause problems for candidates. Most candidates appreciated that $21\text{og}_a 6$ is equal to $\log_a 6^2$ but then a significant number of candidates made the error of equating $\log_a 36 - \log_a 3$ to $\frac{\log_a 36}{\log_a 3}$. Examiners expected to see an intermediate stage between '$\log_a x = \log_a 36 - \log_a 3$' and the printed answer. In part (b), many candidates were able to write the left-hand side of the given equation as a single logarithm but then could not move out of logarithms. A significant number of answers did not involve the constant a or gave the answer as '$5y = 7^a$'. Although the question asked for y in terms of a, examiners condoned a final answer left as $5y = a^7$.

Q3.

This question proved to be a good source of marks for many candidates. Most candidates answered both parts of (a) correctly although the usual errors, $3^\circ = 0$ or $3^\circ = 3$ and $3^X = 27 \Rightarrow x = \sqrt[3]{27}$ were seen. The trapezium rule, required in part (b), was generally well understood although some candidates are still mixing up 'ordinates' and 'strips' or failing to use sufficient brackets correctly. Although many candidates scored well in parts (c) there were others in (i) who used the relevant law of logarithms incorrectly or others who failed to indicate that logarithms had been used at all. In (ii) not all candidates realised that a simple substitution of 13 for 3^X in $k = 27 - 3^X$ with evaluation was all that was required. Descriptions of transformations continue to pose a problem for candidates. Many candidates gained partial credit for their sketches but both marks were rarely awarded.

Q4.

This question proved to be a good source of marks for many of the candidates. Its structure enabled candidates to progress even if they were unsuccessful with some of the parts.

Part (a)(i) Reasonably well done. Errors were with the derivative of e^{2x} Common errors were e^{2x} and $2x\ e^{2x}$ A number of candidates added a ‘+ c’ when they were differentiating.

Part(a)(ii). Usually correct, with errors, if seen, similar to those in the previous part.

Part (b)(i) The majority of candidates answered this correctly.

Part (b)(ii) Very well answered by the majority of candidates.

Part (b)(iii) Again very well answered with many totally correct responses. The major error apart from those who used the incorrect equation $e^{2x} – 5\ e^{2x} + 6$ was the evaluation of e^{21n3} which was often seen as 6 rather than 9. Possibly the evaluation of e^{21n2} as 4 was fortuitous for some candidates.

Part (iv) Similar to part (iii) with similar errors, the major one being the evaluation of $4e^{21n3}$ as 4 × 6 = 24. This resulted in an answer of –6 and hence loss of accuracy marks.

Q5.

Part (a) This was very poorly answered by the majority of candidates. Range is a topic that is not very well understood by candidates.

Part (b) This was reasonably well answered. Errors occurred where candidates, on interchanging x and y, tried to work with $x = 2e^{3x} – 1$.

Part (c) This was poorly answered by most of the candidates. Many candidates obtained the correct differentiation of the In function but lost marks because they omitted to multiply by 1/2. This resulted in the very common incorrect answer of 2/3.

Q6.

Most candidates struggled with part (a) of this question, indicating inexperience, or non understanding, of the formulation of a differential equation from a context situation.

Part (a) Despite the question explicitly requesting a differential equation for $\frac{dx}{dt}$ many candidates just wrote down an expression for x.

A commonly seen differential equation was $dx/dt = kx$ which candidates then proceeded to solve, despite the question stating that a solution of the differential equation was not required. Of those who could translate the given information into correct symbols, a symbol t for time, often found its way into their expression, and then often in an exponential form. Many attempts using the exponential function were seen, suggesting many candidates associating anything to do with differential equations with exponential forms. Those candidates who did give a correct differential equation in part (i) usually found a correct value for the constant of proportionality k. However, there were also many attempts from a partially correct part (i) where the given values of 1000 and 200 were confused, the latter often being taken as a value of time. Other candidates put forward the incorrect argument that the population would be equal to 1200 at time $t = 1$.

Part (b)(i) Virtually all candidates scored at least one mark here; the common error being to round off 5.545 to 5.6 or not to round it at all.

Part (b)(ii) Many candidates solved this equation correctly, although it was not always clear just how they had done it. Most candidates used the technique as per the mark scheme, although some worked successfully from $e^{30} = \left(= \frac{4x}{5000 - x}\right)^4$. Of those who went awry, the error usually involved an incorrect manipulation of the logarithm expression, including ignoring the ln altogether, or stating 4ln 4 = ln l6. Some started correctly with 7.5 = ln $4x$ – ln $(5000 – x)$ but obtained $e^{75} = 4x – (5000 – x)$. Others made relatively simple errors in signs or coefficients in their manipulation towards a solution.

Q7.

In part (a), some candidates did not provide sufficient steps to justify stating the printed value for x. The most common error was illustrated by " $\log_a 216 - \log_a 8 = \frac{\log_a 216}{\log_a 8}$ ".
Most candidates gained significant credit in part (b), although part (b)(iv) did provide some challenge.

Q8.

Many candidates incorrectly evaluated $(3^0 + 1)$ as 1 in part (a) although surprisingly they frequently recovered and used the correct answer, 2, in part (b). In general, the trapezium rule was well understood, although it was relatively common to find an arithmetical slip in evaluating $(3^x + 1)$; the '1' was not added in one of the terms. Candidates should have given the final answer to the degree of accuracy specified in this numerical integration question. Some of the weaker candidates just integrated $(3^x + 1)$ directly as $(3^{x+1} + x)$ The use of a diagram to show that the approximation is an overestimate was well understood, although some used rectangles rather than the required trapeziums and others had the sloping sides of their trapeziums in the wrong positions, that is, below the curve. Part (c) was generally well answered although some weaker candidates tried to take logarithms of each term rather than to rearrange the equation into the form '$3^x = 4$' as a first step. Those candidates who used trial and improvement usually failed to produce an acceptable solution. In part (d), a common wrong expression for f(x) was "$-3^x + 1$", although a significant minority did obtain the correct answer " $3^{-x} + 1$".

Q9.

Part (a) tested the relationship of e^x and ln x. This part was well answered. Part (b) required algebraic manipulation using e^{-x} which discriminated between candidates. The final part was accessible to candidates who failed on the previous part. Candidates were required to give exact answers and not decimal equivalents.